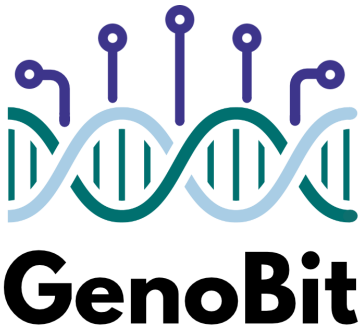


Neuroimaging

Image preprocesing

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1 Introduction

Alzheimer'z disease (AD) is a neurodegenerative disease in elderly people, that causes memory lost and impared thinking. It is characterized by the build up of two main proteins amyloid beta and tau protein that cause abnormal protein clumps called amyloid plaques and tangled fibers known as neurofibrillary tangles in specific areas of the brain, particularly the temporal lobe and neocortical regions (Dash et al., 2023). There are curently around 50 million people worldwide are living with Alzheimer's, and this number is expected to rise to approximately 152 million by 2050 (Ferreira, C. A. et al., 2018). The progress of AD can be classified as three stages: (i) The asymptotic stage, (ii) Mild AD, and (iii) The Dementia stage. In the first stage, changes in the brain begin to occur without any behavioral symptoms. In the second stage, memory complaints may become noticeable to patients (Mahendran and PM, 2022). In the final dementia stage, they are severely impaired and unable to manage their daily lives. There is no known cure, but if diagnosed in an early asymptomatic stage it can be controled. Recent disease-modifying therapies targeting amyloid-beta plaques have shown modest slowing of cognitive decline in early stages. Lecanemab (Cohen et al., 2023) and donanemab (Courade et al., 2025) are two FDA-approved monoclonal antibodies that reduce amyloid plaque and slow down the progression by approximately 27–35% over 18 months in amyloid-positive patients with mild cognitive impairment or mild AD.

2 Oasis directory, image data, structure and file names

Oasis is an opensource directory providing brain imaging data that are freely available for distribution and data analysis. We will be utilizing the Oasis-1 data set this data set consists of a cross-sectional collection of 416 subjects covering the adult life span aged 18 to 96 including individuals with early-stage Alzheimer's Disease (AD).

For each subject, 3 or 4 individual T1-weighted MRI scans obtained within a single imaging session are included. The subjects are all right-handed and include both men and women. 100 of the included subjects over the age of 60 have been diagnosed with very mild to mild AD. Additionally, for 20 of the nondemented subjects, images from a subsequent scan session after a short delay (less than 90 days) are also included as a means of assessing acquisition reliability. All data have been anonymized to accommodate public distribution. Facial features were removed at the fMRIDC (http://www.fmridc.org) using the Brain Extraction Tool. The full set is 15.8 GB compressed and 50 GB uncompressed. The data are available at http://www.oasis-brains.org.

For each subject, a number of images are provided, including: 1) 3-4 images corresponding to multiple repetitions of the same structural protocol within a single session to increase signal-to-noise, 2) an average image that is a motion-corrected coregistered average of all available data, 3) a gain-field corrected atlasregistered image to the 1988 atlas space of Talairach and Tournoux (Buckner et al., 2004), 4) a masked version of the atlas-registered image in which all non-brain voxels have been assigned an intensity value of 0, and 5) a grey/white/CSF segmented image (Zhang et al., 2001). **All images are in 16-bit big-endian Analyze 7.5 format.**

The contents of the DVD and downloadable archive file are organized by imaging session. Data from each MRI session exists in its own directory labeled by the subject ID. The random subject ID uses the format OAS1_xxxx, where 'xxxx' represents a number from 00001 to 9999 (e.g., OAS1_0012). For each of a subject's imaging sessions, an ID has been assigned of the format OAS1_xxxx_MRy,

where y represents an incrementing number to reflect the imaging visit number for the subject (e.g., OAS1_0012_MR1).

For each session, a zip-compressed archive file is included and named with the session ID. When uncompressed, each session directory includes an XML file, a text (TXT) file, and three subdirectories: RAW, PROCESSED, and FSL_SEG.

The **XML file** includes acquisition details and anatomic measures derived from the scan images. A full description of the OASIS XML format can be found at http://www.oasis-brains.org/schemas/.

The **TXT file** includes the same information as the XML but is converted to text format for viewing.

The **RAW directory** includes the individual scan images.

The **PROCESSED directory** includes two additional subdirectories: SUBJ_111 and T88_111. SUBJ_111 includes the averaged, co-registered image of the individual scan images in the native acquisition space in resampled to 1mm isotropic voxels. T88_111 includes the atlas-registered gain field-corrected image and a brain-masked version of that image resampled to 1mm isotropic voxels. It also includes a subdirectory called t4_files that includes the matrices describing the transformation into atlas space.

The **FSL_SEG directory** includes the grey/white/CSF segmentation image generated from the masked atlas image.